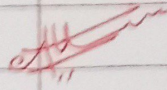
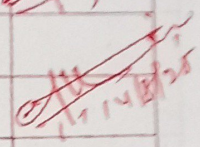
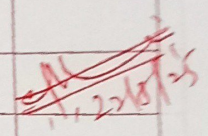
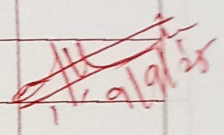
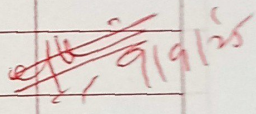
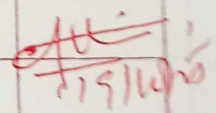
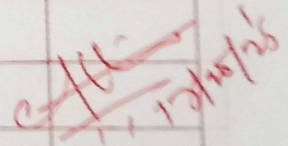
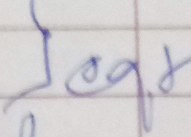
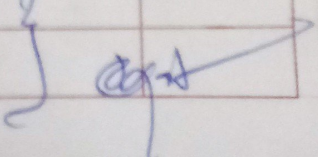


I N D E X

NAME: Kuberan STD.: A1 SEC.: A ROLL NO.: _____ SUB.: Deep learning

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
1	24/7	Exploring Deep learning Platform		
2	31/07	Implement a classifier using open source dataset		
3	14/08	Study of the classifiers with respect to statistical parameters		
4	14/08	Build a simple feed forward neural network to register uc		
5	22/08	Study of Activation function and its role		
6	09/09	Gradient descent & backpropagation in deep neural networks		
7	16/09	Build a cnn model classify cat and dog image		
8	9/10	experiment of LSTM		
9	09/10	Build a Recurrent Neural Network		
10	17/10	Perform Compression on mnist dataset using Autoencoder		
11	17/10	Experiments using variational Autoencoder (VAE)		
12	26/10	Implement a Deep convolutional GAN (DCGAN) to color image		
13	26/10	Understand the Architecture of a pre-trained Model		
14	26/10	Implement a pre-trained CNN model as a Feature Extractor		
15	26/10	Implement a yolo model for object Detection		

26/10/25
Ex: NO. 15

15. Implement a YOLO Model for object Detection

Aim:

To implement and test a YOLO object detection model to locate and classify objects in images.

Objectives

1. To understand real-time object detection
2. To perform bounding box regression and classification
3. To evaluate detection speed and accuracy

Pseudocode

1. start
2. load YOLO model weights and configuration
3. Load input image or video.
4. preprocess image (resize, normalize)
5. perform forward pass to obtain bounding boxes and class probabilities

Output

Detected: person (confidence: 0.96)

Detected: dog (confidence: 0.91)

Detected: bicycle (confidence: 0.87)

Green $\rightarrow$ person

dog $\rightarrow$ blue

Red $\rightarrow$ bicycle.

6. Apply Non-max suppression to remove duplicate boxes

7. Draw bounding boxes and labels on image

8. Display output

End

Observation

1. YOLO provided fast real-time detection with good accuracy

2. Larger objects were detected more reliably than very small objects

3. Bounding Box overlap was reduced using Non-max suppression

Result

Q. YOLO model successfully detected and localized objects in images, demonstrating high-speed object recognition capability.